

Project Proposal — Hotel Booking Cancellation Prediction Using Machine Learning

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1. Certificate Name

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2. Project Title and Description

Title:

Hotel Booking Cancellation Prediction Using Machine Learning

Hotels receive thousands of booking requests, and many reservations are canceled, which affects revenue and resource planning. This project aims to develop a machine learning system that predicts whether a hotel booking will be canceled based on customer details, booking information, and reservation characteristics.

The project will apply supervised learning algorithms to predict booking cancellations and analyze customer booking patterns using machine learning techniques.

3. Problem Type

This project uses:

Learning Type	Problem	Goal
Supervised Learning	Binary Classification	Predict if a booking will be canceled or not
Unsupervised Learning	Clustering	Discover groups of customers with similar booking behaviors

Target Column: is_canceled

Value Meaning

0	Not canceled
1	Canceled

4. Dataset

Dataset Name: Hotel Booking Demand Dataset

Source: Kaggle

<https://www.kaggle.com/dattae/hotel-booking-demand-datasets>

Information Details

Rows	119,390
Columns	32
Data Type	Structured Tabular Data
Target	is_canceled

Main Features:

CATEGORY	EXAMPLES
CUSTOMER DETAILS	adults, children, country, customer type
BOOKING DETAILS	lead_time, stay duration, arrival information
HOTEL DETAILS	hotel, meal, reserved_room_type
PREVIOUS HISTORY	previous_cancellations, previous_bookings_not_canceled
FINANCIAL DATA	adr, deposit_type

5. Data Preprocessing Plan

The following preprocessing steps will be applied:

Step	Description
Data Inspection	Understand dataset structure and data types
Missing Values	Handle missing values in relevant columns
Duplicate Removal	Remove duplicated records
Outlier Handling	Analyze and manage extreme values

Step	Description
Encoding	Apply One-Hot Encoding for categorical variables
Scaling	Scale numerical features
Data Splitting	Split data into training and testing sets (80/20)

Encoding Method: One-Hot Encoding

One-Hot Encoding is selected because most categorical variables do not have a natural order.

6. Algorithms I Plan to Train

The following classification algorithms will be trained and compared:

Algorithm	Reason
Logistic Regression	Baseline model for binary classification and easy interpretation
Random Forest Classifier	Handles complex relationships and reduces overfitting
Gradient Boosting Classifier	Strong ensemble algorithm for tabular datasets

For customer behavior analysis, clustering algorithms will also be applied:

Algorithm	Purpose
K-Means Clustering	Groups customers based on similarities
Agglomerative Clustering	Creates hierarchical customer groups

7. Evaluation Plan

Classification Metrics:

Metric	Purpose
Accuracy	Overall prediction performance

Metric	Purpose
Precision	Correct cancellation predictions
Recall	Detecting actual cancellations
F1-Score	Balance between precision and recall
Confusion Matrix	Shows prediction errors
ROC-AUC	Measures model performance

The best model will be selected based on F1-score.

Clustering Evaluation:

- Elbow Method
- Silhouette Score

8. Deployment Plan

The best trained model can be deployed using **FastAPI**.

The API will accept booking information and return a cancellation prediction with probability.

Example:

Input:

```
{
  "hotel": "City Hotel",
  "lead_time": 120,
  "adults": 2,
  "adr": 150
}
```

Output:

```
{
  "prediction": "Canceled", "probability": 0.82
}
```

9. Repository Plan

hotel-booking-ml-project/

├── dataset/

| └── hotel_bookings.csv

├── notebooks/

| ├── 01_data_cleaning_eda.ipynb

| └── 02_model_training.ipynb

├── src/

| ├── preprocess.py

| └── train.py

├── models/

| └── best_model.pkl

├── reports/

| └── project_report.pdf

├── README.md

├── project_proposal.md

└── requirements.txt

Conclusion

This project applies supervised and unsupervised machine learning techniques to real-world hotel booking data. The classification models will predict booking cancellations, while clustering methods will provide additional insights into customer booking patterns.

The project demonstrates skills in data preprocessing, exploratory data analysis, machine learning model development, evaluation, and deployment.